

Practical 7:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 10;
static int static_var = 20;

void code_function()
{
    printf("Inside code function\n");
}

int main()
{
    int stack_var = 30;
    static int static_local = 40;
    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL)
    {
        perror("malloc");
        return 1;
    }

    *heap_var = 50;

    printf("PID                : %d\n", getpid());

    printf("\nMemory Addresses:\n");
    printf("Code (function)          : %p\n", (void *)code_function);
    printf("Global variable          : %p\n", (void *)&global_var);
    printf("Static variable          : %p\n", (void *)&static_var);
    printf("Static local             : %p\n", (void *)&static_local);
    printf("Heap variable            : %p\n", (void *)heap_var);
    printf("Stack variable           : %p\n", (void *)&stack_var);

    printf("\nProcess is running. Press Enter to exit...\n");

    printf("\nProcess is running. Press Enter to exit...\n");
    getchar();

    free(heap_var);

    return 0;
}
```

```
bhanu@DESKTOP-DE32HQ8:~$ mkdir -p ~/memory_analysis
cd ~/memory_analysis
bhanu@DESKTOP-DE32HQ8:~/memory_analysis$ pwd
/home/bhanu/memory_analysis
bhanu@DESKTOP-DE32HQ8:~/memory_analysis$ nano memory.c
bhanu@DESKTOP-DE32HQ8:~/memory_analysis$ gcc -Wall -Wextra -o memory memory.c
bhanu@DESKTOP-DE32HQ8:~/memory_analysis$ ls
memory  memory.c
bhanu@DESKTOP-DE32HQ8:~/memory_analysis$ ./memory
PID                : 745

Memory Addresses:
Code (function)    : 0x57433bdf5229
Global variable    : 0x57433bdf8010
Static variable    : 0x57433bdf8014
Static local       : 0x57433bdf8018
Heap variable      : 0x57435b15c2a0
Stack variable     : 0x7fff1fef999c

Process is running. Press Enter to exit...

bhanu@DESKTOP-DE32HQ8:~/memory_analysis$ |
```